

Angel Santoyo

Goal: Use the AWS Pricing Calculator to estimate monthly costs and compare multiple AWS configurations (EC2, scaling, serverless, and database) using a California-based region. You will create 4 configurations and compare them:

- EC2 (t3.micro) baseline
- EC2 scaled (t3.medium ×2)
- Serverless alternative (Lambda)

Part 1: Access the AWS Pricing Calculator

- Open a browser and go to: <https://calculator.aws>
- Click “Create estimate”
- You do not need to log in to AWS to use the calculator.

The screenshot shows the AWS Pricing Calculator interface. At the top, there's a navigation bar with 'aws pricing calculator', 'Feedback', 'Language: English', 'Contact Sales', and 'Create an AWS Account'. Below this, the main content area is titled 'Add service' and includes a 'Bulk import' button. The interface is divided into two main sections: 'Step 1: Add service' and 'Step 2: Configure service'. In the 'Add service' section, there are two search options: 'Search by location type' (selected) and 'Search all services'. The 'Search by location type' section includes a 'Choose a location type' dropdown set to 'Region' and a 'Choose a Region' dropdown set to 'US East (Ohio)'. Below these is a 'Find Service' search bar. The 'Search all services' section is currently empty. The 'Configure service' section displays three service cards: 'AWS Application Migration Service', 'Amazon API Gateway', and 'Amazon AppFlow'. Each card includes a brief description and a 'Configure' button. At the bottom of the interface, there's a summary bar showing 'Upfront cost: 0.00 USD', 'Monthly cost: 0.00 USD', and 'Total 12 months cost: 0.00 USD (includes upfront cost)', along with a 'View summary' button.

Part 2: Configuration 1 – Basic EC2 Web Server

Scenario: A small web application running 24/7.

- Click Add service
- Select Amazon EC2. Under EC2 Instance:
 - Region: US West (Northern California)
 - Operating system: Linux
 - Instance family: t3
 - Instance type: t3.micro
 - Tenancy: Shared

- Number of instances: 1
- Usage: 730 hours/month

Edit Amazon EC2 [info](#)

Description
Enter a description for your estimate

Choose a location type [info](#) Choose a Region
 Region ▼ US West (N. California) ▼

EC2 specifications [info](#)

Tenancy
Choose the tenancy type to run your Amazon EC2 instances on.
 Shared Instances ▼

Operating system
Choose the operating system to run your Amazon EC2 instances on.
 Linux ▼

Workloads
Choose the graph that best represents your monthly workload
☒ Constant usage ☐ Daily spike traffic ☐ Weekly spike traffic ☐ Monthly spike traffic

Number of instances
Please specify the total number of instances that you need each month.
 1 ↕

EC2 Instances (596)
 Based on your inputs, this is the lowest-cost EC2 instance: **t4g.nano**
 Chosen instance: **t3.micro** | Family: **t3** | 2vCPU | 1 GiB Memory
 Search instance type

Instance family [info](#) vCPUs Memory (GiB) Network performance

Total Upfront cost: 0.00 USD Show Details ▼ Cancel Update

Total Monthlv cost: 11.93 USD

- Storage:
 - EBS volume type: General Purpose SSD (gp3)
 - Storage size: 30 GB
- Data transfer:
 - Outbound data: 10 GB/month
- Click Save and add service
- Record the following:
 - Monthly estimated cost
 - Which components drive most of the cost

Monthly cost is driven a lot by ec2 compute type, storage and data transfer.

Operating system
Choose the operating system to run your Amazon EC2 instances on.
 Linux ▼

Workloads
Choose the graph that best represents your monthly workload
☒ Constant usage ☐ Daily spike traffic ☐ Weekly spike traffic ☐ Monthly spike traffic

Number of instances
Please specify the total number of instances that you need each month.
 2 ↕

EC2 Instances (596)
 Based on your inputs, this is the lowest-cost EC2 instance: **t4g.nano**
 Chosen instance: **t3.medium** | Family: **t3** | 2vCPU | 4 GiB Memory
 Search instance type

Instance family [info](#) vCPUs Memory (GiB) Network performance

Any Instance family ▼ Any vCPUs ▼ Any Memory (GiB) ▼ Any Network Performance ▼

☒ Show only current generation instances.

< 1 2 3 4 5 6 7 ... 60 > [info](#)

	Instance name	vCPUs	Memory	Network Performance	Storage	On-Demand Hourly Cost	CurrentGeneration	Potential Effective Hourly Cost (Savings %)
☐	t4g.nano	2	0.5 GiB	Up to 5 Gigabit	EBS only	0.005	Yes	0.0019 (63%)
☐	t3a.nano	2	0.5 GiB	Up to 5 Gigabit	EBS only	0.0056	Yes	0.0021 (63%)
☐	t3.nano	2	0.5 GiB	Up to 5 Gigabit	EBS only	0.0062	Yes	0.0023 (63%)
☐	t7.nano	1	0.5 GiB	1 net	FBS only	0.0069	Yes	0.0030 (56%)

Estimated commitment price based on the following selections:
Instance type: **t3.medium** Operating system: **Linux**

Select the container and options to find your best price

☐ Compute Savings Plans
One plan that automatically applies to all usage on EC2, Fargate, and Lambda. Up to 66% discount. [Learn more](#)
Reservation term
☐ 1 year
☒ 3 year
Payment Options
☒ No upfront
☐ Partial upfront
☐ All upfront

Upfront: 0.00
Monthly: 20.81/Month

☐ EC2 Instance Savings Plans
Get deeper discount when you only need one instance family and region. Up to 72% discount. [Learn more](#)
Reservation term
☐ 1 year
☒ 3 year
Payment Options
☒ No upfront
☐ Partial upfront
☐ All upfront

Upfront: 0.00
Monthly: 15.62/Month

☒ On-Demand
Maximize flexibility. [Learn more](#)
Expected utilization
Enter the expected usage of Amazon EC2 instances
Usage
730
Usage type
Hours / Month

Note that for usage less than 100%, the calculator assumes that the attached EBS volume is deleted while the instance is not running. Keep in mind that volumes are billed as long as they are provisioned in your account, even while attached to a stopped instance.

Instance: 0.0496/Hour
Monthly: 36.21/Month

☐ Spot Instances
Minimize cost by leveraging EC2's spare capacity. Recommended for fault tolerant and interruption tolerant applications. [Learn more](#)
The historical average discount for t3.medium is 59%
Assume percentage discount for my estimate
59

Actual spot instance pricing varies
With spot instances, you pay the spot price that's in effect for the time period your instance is running

Instance: 0.0496/Hour
Monthly: 29.69/Month

► Other purchasing options

► Show calculations

Part 3: Configuration 2 – Scaled Web Application

Scenario: A production workload with redundancy and higher traffic. Implement the following changes.

- Duplicate the EC2 service (or add a new one)
- Modify the new EC2 service:
 - Instance type: t3.medium
 - Number of instances: 2
 - Usage: 730 hours/month

Instance type	Count	Memory	Up to 2 vcpus	EBS only	Monthly	Yes	Monthly (62%)
☒ t3.medium	2	4 GiB	Up to 5 Gigabit	EBS only	0.0416	Yes	0.0156 (62%)
☐ t3a.large	2	8 GiB	Up to 5 Gigabit	EBS only	0.0752	Yes	0.0283 (62%)
☐ t3.large	2	8 GiB	Up to 5 Gigabit	EBS only	0.0832	Yes	0.0313 (62%)

Payment options

Estimated commitment price based on the following selections:
Instance type: **t3.medium** Operating system: **Linux**

Select the container and options to find your best price

☒ Compute Savings Plans
One plan that automatically applies to all usage on EC2, Fargate, and Lambda. Up to 66% discount. [Learn more](#)
Reservation term
☐ 1 year
☒ 3 year
Payment Options
☒ No upfront
☐ Partial upfront
☐ All upfront

Upfront: 0.00
Monthly: 15.04/Month

☐ EC2 Instance Savings Plans
Get deeper discount when you only need one instance family and region. Up to 72% discount. [Learn more](#)
Reservation term
☐ 1 year
☒ 3 year
Payment Options
☒ No upfront
☐ Partial upfront
☐ All upfront

Upfront: 0.00
Monthly: 13.14/Month

☐ On-Demand
Maximize flexibility. [Learn more](#)
Expected utilization
Enter the expected usage of Amazon EC2 instances
Usage
730
Usage type
Hours / Month

Note that for usage less than 100%, the calculator assumes that the attached EBS volume is deleted while the instance is not running. Keep in mind that volumes are billed as long as they are provisioned in your account, even while attached to a stopped instance.

Instance: 0.0416/Hour
Monthly: 30.37/Month

☐ Spot Instances
Minimize cost by leveraging EC2's spare capacity. Recommended for fault tolerant and interruption tolerant applications. [Learn more](#)
The historical average discount for t3.medium is 52%
Assume percentage discount for my estimate
-1

Actual spot instance pricing varies
With spot instances, you pay the spot price that's in effect for the time period your instance is running

Instance: 0.0416/Hour
Monthly: 30.67/Month

3. Storage:

- EBS volume type: General Purpose SSD (gp3)

- Storage size: 50 GB per instance
- Data transfer:
 - Outbound data: 100 GB/month

General Purpose SSD (gp3) - IOPS
gp3 supports a max of 80,000 IOPS per volume
Enter amount of IOPS per volume

General Purpose SSD (gp3) - Throughput
gp3 supports a max of 2000 MBps per volume

Value Unit

Storage amount Unit

Snapshot Frequency
No snapshot storage

► Show calculations

► Detailed monitoring - *optional* [info](#)

▼ **Data transfer - optional**

Inbound Data Transfer
Enter the data you expect to transfer into US West (Oregon)

Data transfer from Enter Amount Data amount

Intra-Region Data Transfer
Enter the data you expect to transfer between Availability Zones or VPC Peering connections in US West (Oregon)

Enter Amount Data amount

Outbound Data Transfer
Enter the data you expect to transfer out of US West (Oregon)

Data transfer to Enter Amount Data amount

- Save the configuration

[AWS Pricing Calculator](#) > My Estimate

My Estimate [Edit](#) [info](#)

Estimate summary [info](#)

Upfront cost 0.00 USD	Monthly cost 29.03 USD	Total 12 months cost 348.36 USD Includes upfront cost
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[Get started](#)

My Estimate [info](#)

☐	Service Name	Status	Upfront cost	Monthly cost	Description
☐	Amazon EC2	-	0.00 USD	9.99 USD	-
☒	Amazon EC2	-	0.00 USD	19.04 USD	-

Acknowledgement

Part 4: Configuration 3 – EC2 vs Serverless (Lambda)

Scenario: Replace always-on servers with event-driven compute.

- Click Add service
- Select AWS Lambda

- Configure:
 - Requests: 1 million per month
 - Duration: 200 ms
 - Memory: 512 MB

Create estimate: Configure AWS Lambda

Description
Enter a description for your estimate

Choose a location type
Region: [Dropdown]

Choose a Region
US West (Oregon)

Service settings

Architecture
x86

Number of requests
1000000 per month

Duration of each request (in ms)
200

Amount of memory allocated
512 MB

Total Upfront cost: 0.00 USD
Total Monthly cost: 0.00 USD

Buttons: Cancel, Save and view summary, Save and add service

- Save the configuration

Estimate summary

Upfront cost: 0.00 USD
Monthly cost: 29.05 USD
Total 12 months cost: 348.36 USD (Includes upfront cost)

My Estimate

Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
Amazon EC2	-	0.00 USD	9.99 USD	-	US West (Oregon)	Tenancy (Shared Instances), Operat...
Amazon EC2	-	0.00 USD	19.04 USD	-	US West (Oregon)	Tenancy (Shared Instances), Operat...
AWS Lambda	-	0.00 USD	0.00 USD	-	US West (Oregon)	Architecture (x86), Architecture (x86...

Buttons: Export, Share, Duplicate, Delete, Move to, Create group, Add support, Add service

This AWS Lambda appears to be using the free tier with zero cost.

Questions:

Part 5: Add a Managed Database (RDS)

Scenario: Add a backend database.

- Click Add service
- Select Amazon RDS
- Configure:
 - Engine: MySQL
 - Instance type: db.t3.micro

- Deployment: Single-AZ
- Storage: 20 GB

Selected Instance:
db.t3.micro
vCPU: 2
Memory: 1 GiB

Utilization (On-Demand only)
With utilization, you still have to stop the instance to get the cost benefit. Utilization only affects OnDemand pricing for instances and not the storage, backups, etc.

Value

100

Unit

%Utilized/Month

Deployment option

Single-AZ

Pricing model

OnDemand

Show calculations

RDS Proxy

Info

Would you be creating an RDS Proxy with the database?

Yes

Show calculations

Storage

Info

Storage for each RDS instance

General Purpose SSD (gp2)

Storage amount

20

Unit

GB

- Save

AWS Pricing Calculator > My Estimate

My Estimate [Edit](#)

Estimate summary

Upfront cost

0.00 USD

Monthly cost

154.38 USD

Total 12 months cost

1,852.56 USD

Includes upfront cost

Getting Started

Get started

My Estimate

Find resources

☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region
☐	Amazon EC2	-	0.00 USD	11.93 USD	-	US We
☐	Amazon EC2	-	0.00 USD	82.02 USD	-	US We
☐	AWS Lambda	-	0.00 USD	0.00 USD	-	US We
☐	Amazon RDS for MySQL	-	0.00 USD	60.43 USD	-	US We

Duplicate

Delete

My Estimate

Deliverable:

- Make a cost comparison table

Configuration	Monthly Cost	Primary Cost Driver
EC2 t3.micro (Baseline)	\$11.93/month	EC2 compute (instance hours)
EC2 t3.medium ×2	\$82.02/month	Larger instance + multiple instances
AWS Lambda (1M requests)	~\$0 (Free Tier)	Execution time (Free Tier covers usage)
Amazon RDS	\$60.43/month	Database instance compute

- Compare the difference between the cost of configuration 1 vs. configuration 2. What is the biggest cost factor?

Configuration 1 costs \$11.93 per month, while Configuration 2 costs \$82.02 per month. The difference between them is \$70.09 per month. The biggest cost factor is EC2 compute scaling. Configuration 2 increases both the instance size (from t3.micro to t3.medium) and the number of instances (from 1 to 2), while still running 730 hours per month. Because EC2 charges based on instance type and uptime, these changes significantly increase the total cost.

- Compare the Lambda monthly cost with configuration 1.

AWS Lambda costs \$0.00 per month in this scenario because the workload falls within the Free Tier limits. In comparison, Configuration 1 (EC2 t3.micro) costs \$11.93 per month. Lambda is significantly cheaper because it only charges for execution time and does not incur costs for idle compute.

- When is serverless more cost-effective?

Serverless is more cost-effective when workloads are event driven, intermittent, or unpredictable. Because AWS Lambda charges only for the time the function runs, it avoids paying for always on infrastructure. It is ideal for applications with low to moderate traffic or burst-based usage patterns.

Write a short reflection on what surprised you about AWS pricing.

One surprising aspect of AWS pricing is how quickly costs increase when scaling compute resources. Even small changes in instance size and count noticeably impact monthly expenses. Another notable observation is how inexpensive serverless computing can be at low usage levels, especially when covered by the Free Tier. Also, managed services like RDS can represent a significant portion of total costs, sometimes exceeding the application server itself. This shows how architectural choices directly influence cloud spending.